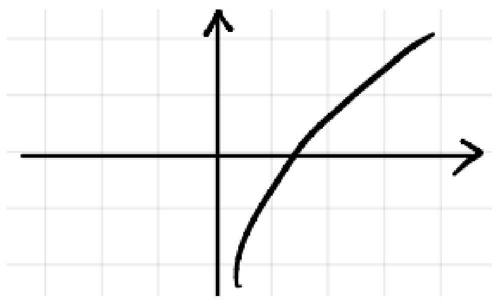
5. Complex Logarithm Function

$$\ln z \triangleq \ln re^{i\theta} = \ln r + i\theta \neq \ln(x+iy)$$

With $\ln(x+iy)$, we can do nothing about it, so we switch to $\ln re^{i\theta}$.

Example 5

$$\begin{aligned} \ln i &= \ln e^{i\frac{\pi}{2}} = i\frac{\pi}{2} \\ \ln\left(\frac{1+i}{1-i}\right) &= \ln(1+i) - \ln(1-i) \\ &= \ln\sqrt{2} + i\frac{\pi}{4} - \left(\ln\sqrt{2} + i\frac{-\pi}{4}\right) = i\frac{\pi}{2} \end{aligned}$$



They have the same value. The function of a real variable $\ln(x)$ is monotonic, but the situation is different for the function of a complex variable $\ln(z)$, which is a multivalued function.

$$\ln z = \ln re^{i\theta} = \ln re^{i(\theta+2\pi)} = \ln re^{i(\theta+2k\pi)}$$

$$e^{i\theta} = e^{i(\theta+2\pi)} = e^{i(\theta+2k\pi)} = \ln r + i(\theta+2k\pi), \quad k = 0, \pm 1, \pm 2$$

However, the following functions are not multivalued:

$$e^z = e^{x+iy} \quad e^z = e^{re^{i(\theta+2k\pi)}}$$

$$\sin z, \quad \cos z, \quad z^n$$

Example

$$\begin{aligned} z^2 &= \left(re^{i\theta}\right)^2 = \left[re^{i(\theta+2k\pi)}\right]^2 \\ &= r^2e^{i(2\theta+4k\pi)} = r^2e^{i2\theta} \text{ not depend on } k \end{aligned}$$

z^n is not a multivalued function, but if a is not an integer, z^a is a multivalued function.

$$
\begin{aligned}
z^{1/2} &= \left[re^{i(\theta+2k\pi)}\right]^{\frac{1}{2}} = r^{\frac{1}{2}}e^{i(\theta/2+k\pi)} = r^{1/2}\theta^{i\theta/2}\cdot e^{ik\pi} \\
&= \begin{cases} r^{1/2}e^{-i\theta/2} & k = \text{even integer} \\ -r^{1/2} & k = \text{odd integer} \end{cases}
\end{aligned}
$$

Suppose x and y are real numbers. Then we can generalize it to the complex domain.

$$x^y = e^{y\ln x}$$

$$\Rightarrow z_1^{z_2} = e^{z_2 \ln z_1}$$

Example

$$i^i = e^{i\ln i} = e^{i\left(i\frac{\pi}{2}+2k\pi\right)} = e^{-\frac{\pi}{2}-2k\pi}$$

Example

$$
\begin{aligned}
(-1)^{\sin i} &= (-1)^{\frac{e^{-1}-e^{1}}{2i}} \\
&= e^{\frac{e^{-1}-e^{1}}{2i}\ln(-1)} \\
&= e^{\frac{e^{-1}-e^{1}}{2i}+i(\pi+2k\pi)} \\
&= e^{-\frac{e^{-1}-e^{1}}{2}\cdot(\pi+2k\pi)}
\end{aligned}
$$

Chapter 2 Analytic Function

Here are some complex functions:

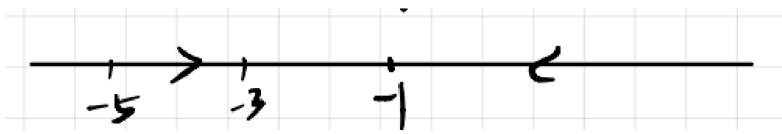
$$z+1,\quad z^2,\quad 2z^2+z-1,\quad e^z,\quad e^{5z^2+2}$$

$$\cos z,\quad \cos(2z^2+z-1),\quad \ln z,\quad z_1^{z_2},\quad \frac{P(z)}{Q(z)},\quad \cosh z,\quad \sinh z$$

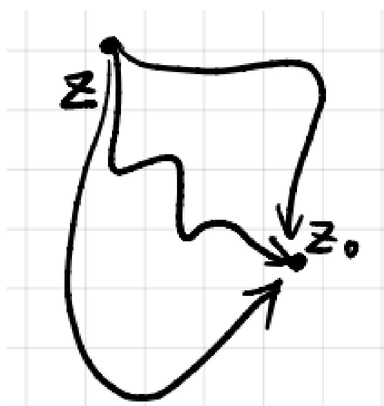
2.1 Limit

$$f(x)\quad x\to x_0\quad |x-x_0|\to 0$$

On the real domain, the limit is approximated from two or one direction.

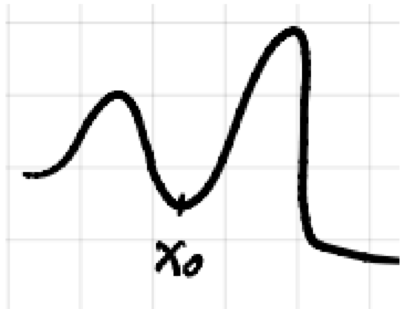


On the complex domain (2-D), the limit is approximated from multiple directions.



2.2 Continuous function

$$x\to x_0$$
$$f(x)\to f(x_0)$$



Then we can generalize it to the complex domain. If a function is continuous at point z_0, then $f(z) \to f(z_0)$ when $z \to z_0$.

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Definition: the limit of continuous complex functions:

If $|f(z) - f(z_0)| \to 0$, when $|z - z_0| \to 0$.

$$\lim_{z \to z_0} f(z) = f(z_0)$$

3. Derivative

Real function

A real function $f(x)$ has a derivative at x:

$$f'(x) \triangleq \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Example:

$$\lim_{\Delta x \to 0} \frac{(x_1 + \Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \to 0} \frac{x^2 + 2x \cdot \Delta x + \Delta x^2 - x^2}{\Delta x} = \lim_{\Delta x \to 0} (2x + \Delta x) = 2x$$

Now, let's move to complex functions. The derivative of $f(z)$ at z_0 is:

$$f'(z) \triangleq \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Example 1:

$$f(z) = z^2$$

$$f'(z) = \frac{z^2 + 2z \cdot \Delta z + (\Delta z)^2 - z^2}{\Delta z} = 2z \cdot \Delta z + (\Delta z)^2$$

Example 2:

$$f(z) = z^2, \quad z = x + iy$$

$$f(z) = (x + iy)^2 = x^2 - y^2 + i(2xy) = u(x, y) + iv(x, y)$$

Example 3:

$$\begin{aligned} f(z) \quad &= \frac{1 + z}{1 - z} = \frac{1 + x + iy}{1 - x - iy} \\ &= \frac{(1 + x + iy)(1 - x + iy)}{(1 - x)^2 + y^2} = \frac{(1 + iy)^2 - x^2}{(1 - x)^2 + y^2} \\ &= \frac{1 - y^2 + i2y - x^2}{(1 - x)^2 + y^2} \end{aligned}$$

Thus, we have:

$$u(x, y) = \frac{1 - y^2 - x^2}{(1 - x)^2 + y^2}, \quad v(x, y) = \frac{2y}{(1 - x)^2 + y^2}$$

With the above knowledge, we can easily solve the following equations:

$$\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = \pi, \quad \int_{-\infty}^{+\infty} \frac{\sin^4 x}{x^4} = \frac{2\pi}{3}$$

$$\int_0^{\infty} \frac{\ln x}{1 + x + x^2} dx, \quad \sum_{i=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

4.Cauchy-Riemann condition

A function is analytic if it satisfies the Cauchy-Riemann conditions:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$